

1.

A: This is the input array: [38, 21, 39, 60, -1, 10, 81, 23]

sumCounter = 0

[38, 21, 39, 60]

[38 + 21] = 59

sumCounter = 1

[39 + 60] = 99

sumCounter = 2

[-1, 10, 81, 23]

[-1 + 10] = 9

sumCounter = 3

[81 + 23] = 104

sumCounter = 4

59 + 99 = 158

sumCounter = 5

9 + 104 = 113

sumCounter = 6

158 + 113 = 271

sumCounter = 7

B: This is the input array: [2, 97, 5, 88, 9, 72, 12, 64, 17, 56, 21]

sumCounter = 0

[2, 97, 5, 88, 9, 72]

[2 + 97] = 99

sumCounter = 1

[5 + 88] = 93

sumCounter = 2

[9 + 72] = 81

sumCounter = 3

[12, 64, 17, 56, 21]

[12 + 64] = 76

sumCounter = 4

[17 + 56] = 73

sumCounter = 5

[21] = 21

[99, 93, 81, 76, 73, 21]

[99 + 93] = 192

sumCounter = 6

[81 + 76] = 157

sumCounter = 7

[73 + 21] = 94

sumCounter = 8

[192, 157, 94]

[192+157] = 349

sumCounter = 9

349 + 94 = 443

sumCounter = 10

C: This is the input array: [100, 33, 22, 213, 65, 29, 153, 199, 47, 181, 85]

[100, 33, 22, 213, 65, 29]

sumCounter = 0

[100 + 33] = 133

sumCounter = 1

[22 + 213] = 235

sumCounter = 2

[65 + 29] = 94

sumCounter = 3

[153, 199, 47, 181, 85]

[153 + 199] = 352

sumCounter = 4

[47 + 181] = 228

sumCounter = 5

[85] = 85

[133, 235, 94, 352, 228, 85]

[133, 235, 94]

[133 + 235] = 368

sumCounter = 6

[94] = 94

[352, 228, 85]

[352 + 228] = 580

sumCounter = 7

[85] = 85

[368, 94, 580, 85]

[368 + 94] = 462

sumCounter = 8

[580 + 85] = 665

sumCounter = 9

462 + 665 = 1127

sumCounter = 10

2.

A: This is the input array: [38, 21, 39, 60, -1, 10, 81, 23]

Left: [38] Right: [21]

Merged: [21, 38]

Left: [39] Right: [60]

Merged: [39, 60]

Left: [21, 38] Right: [39, 60]

Merged: [21, 38, 39, 60]

Left: [-1] Right: [10]

Merged: [-1, 10]

Left: [81] Right: [23]

Merged: [23, 81]

Left: [-1, 10] Right: [23, 81]

Merged: [-1, 10, 23, 81]

Left: [21, 38, 39, 60] Right: [-1, 10, 23, 81]

Merged: [-1, 10, 21, 23, 38, 39, 60, 81]

Sorted array: [-1, 10, 21, 23, 38, 39, 60, 81].

B: This is the input array: [2, 97, 5, 88, 9, 72, 12, 64, 17, 56, 21]

Left: [2] Right: [97]

Merged: [2, 97]

Left: [88] Right: [9]

Merged: [9, 88]

Left: [5] Right: [9, 88]

Merged: [5, 9, 88]

Left: [2, 97] Right: [5, 9, 88]

Merged: [2, 5, 9, 88, 97]

Left: [12] Right: [64]

Merged: [12, 64]

Left: [72] Right: [12, 64]

Merged: [12, 64, 72]

Left: [56] Right: [21]

Merged: [21, 56]

Left: [17] Right: [21, 56]

Merged: [17, 21, 56]

Left: [12, 64, 72] Right: [17, 21, 56]

Merged: [12, 17, 21, 56, 64, 72]

Left: [2, 5, 9, 88, 97] Right: [12, 17, 21, 56, 64, 72]

Merged: [2, 5, 9, 12, 17, 21, 56, 64, 72, 88, 97]

Sorted array: [2, 5, 9, 12, 17, 21, 56, 64, 72, 88, 97]

C: This is the input array: [100, 33, 22, 213, 65, 29, 153, 199, 47, 181, 85]

Left: [100] Right: [33]

Merged: [33, 100]

Left: [213] Right: [65]

Merged: [65, 213]

Left: [22] Right: [65, 213]

Merged: [22, 65, 213]

Left: [33, 100] Right: [22, 65, 213]

Merged: [22, 33, 65, 100, 213]

Left: [153] Right: [199]

Merged: [153, 199]

Left: [29] Right: [153, 199]

Merged: [29, 153, 199]

Left: [181] Right: [85]

Merged: [85, 181]

Left: [47] Right: [85, 181]

Merged: [47, 85, 181]

Left: [29, 153, 199] Right: [47, 85, 181]

Merged: [29, 47, 85, 153, 181, 199]

Left: [22, 33, 65, 100, 213] Right: [29, 47, 85, 153, 181, 199]

Merged: [22, 29, 33, 47, 65, 85, 100, 153, 181, 199, 213]

Sorted array: [22, 29, 33, 47, 65, 85, 100, 153, 181, 199, 213]